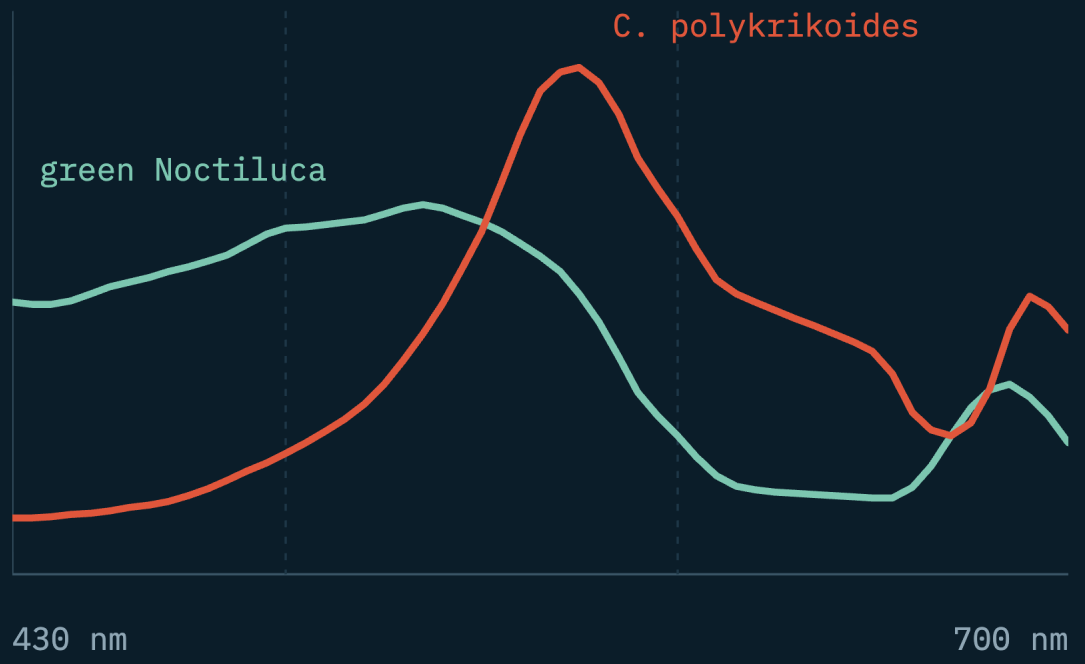
ARAB YOUTH SPACE HACKATHON 2026-CHALLENGE 813

Distinguishing Dangerous Algal Species From Harmless Ones Via Hyperspectral Imagery



Team FireFlies
Tibyan Khaled Ahmed Bilal
MHD Manar Mallouhi

Theme: Water Quality & Inland/Coastal Water Intelligence
Country: Sudan & Syria

01 · Problem

A chlorophyll map shows a bloom, not its species

1,200 km

Of coastline was hit by the 2008 to 2009 *C. polykrikoides* red tide's catastrophic incident. Fish died in large numbers and desalination plants were shut for months.

Richlen et al. 2010

270 of 436

Recorded blooms in the northern Gulf and Sea of Oman between 1980 and 2015 were ***C. polykrikoides***.

Mirza Esmaeili et al. 2021

Each winter

Green *Noctiluca* spreads across the Sea of Oman. It is not toxic, yet on a chlorophyll map it looks just as alarming.

NASA Earth Observatory

Operators get the same warning for both, but they need to know is which one is moving toward their intake.

02 · Area and Timeframe

We studied the Gulf and Sea of Oman over 18 months



Area of interest

The Arabian Gulf, Strait of Hormuz and Sea of Oman on a 0.01° grid, about 1.1 km.

Timeframe

October 2024 to March 2026, the PACE era.
12 scenes: 9 bloom dates, 2 calm dates and 1 held-out test date.

Why here

Cities on this coast drink desalinated seawater, and both species bloom in these waters.

03 · Prior Work

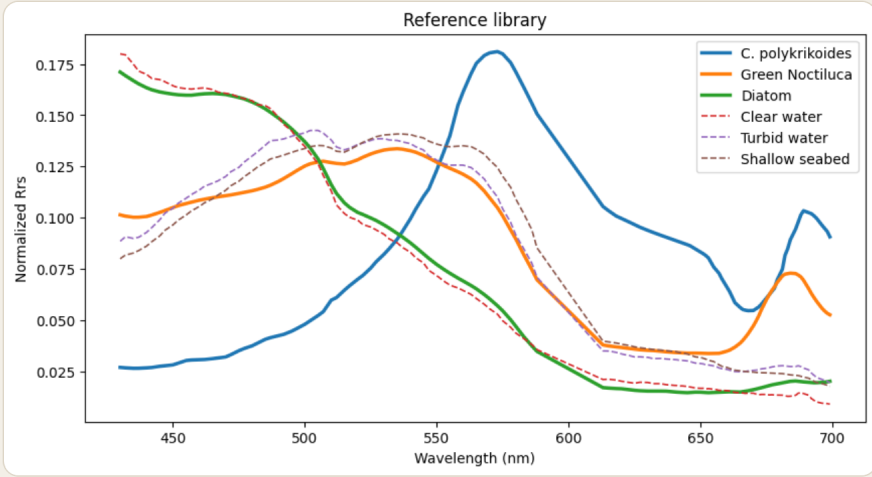
Earlier work finds blooms, not species

Work	Data	What it showed	Still open
Zhao and Ghedira 2014; Moradi and Kabiri 2012	MODIS	Tracked the 2008 red tide with fluorescence	Bloom or no bloom only
Zhao, Temimi and Ghedira 2015	MERIS	Fluorescence rules for blooms in the Gulf and Sea of Oman	Thresholds tied to a sensor retired in 2012
Kim, Yoo and Son 2016; Dwivedi et al. 2015	Field spectra	Reflectance shapes of C. polykrikoides, green Noctiluca and diatoms	Measured in Korea and the Arabian Sea, not tested on PACE
LaHaye, Luis and Gierach 2026	PACE with multispectral	Self-supervised bloom mapping	Built for US coastlines
Satelytics, KSAT	Commercial	Bloom alerts as a service	No published species split

04 · Multispectral vs Hyperspectral

Hyperspectral data keeps the shape of the spectrum

Type	Multispectral	Hyperspectral (PACE OCI)
Visible bands	About 10 (MODIS, OLCI)	172, from 346 to 719 nm
Band spacing	Uneven, gaps up to about 90 nm	Every 2.5 nm
What you measure	Band ratios, chlorophyll, fluorescence height	Full curve shape, peak position, derivatives
Telling species apart	Local thresholds tuned to one sensor	Shape matching against reference spectra



C. polykrikoides peaks near 570 nm, green *Noctiluca* near 530 nm. At full resolution they are 28.7° apart in spectral angle.

A handful of bands only samples a few points on these curves, so two blooms with the same chlorophyll can look identical.

05 · Our Solution

A species-level risk map from every PACE pass

Each bloom pixel gets a likely species, a risk tier and a confidence score.

- **High** · *C. polykrikoides*-like
- **Watch** · green *Noctiluca*-like
- **Watch** · dense bloom, unclassified
- **Monitor** · weak bloom, unclassified
- **Low** · diatom-like

Species, not only blooms

Separates red tide, green *Noctiluca* and diatom-like water.

No labelled datasets needed

PCA and clustering learn patterns from the scenes. References only name them.

Honest when unsure

A name needs a clear margin and two spectral comparison tests that agree. Otherwise: unclassified.

Built to act on

Our iteractive web map.

06 · Who Benefits

Built for people who have to act before the bloom arrives

Desalination operators

UAE east coast and Oman. Notice to prepare intakes and pre-treatment.

Aquaculture and fisheries

Time to protect or move cages before a fish-killing bloom.

Environment Agency Abu Dhabi

Send sampling boats to the pixels where species is uncertain.

UAE Ministry of Climate Change and Environment

National bloom bulletins with species context.

Oman fisheries authorities

Seasonal Noctiluca tracking on the Sea of Oman side.

ROPME and researchers

One view across borders, built only on open data.

06 · Business Use Case

From a satellite pass to a decision at the intake

END USER

Water quality lead at a desalination plant

UAE east coast and Oman. Needs early warning, and to know whether a bloom is the dangerous kind before changing operations.

01 PACE passes over the region every day.

02 The pipeline labels every bloom pixel with species, risk tier and confidence.

03 The plant gets an alert only when a Watch or High bloom sits within its zone.

04 Staff sample the water, prepare pre-treatment, or plan an intake shutdown.

Site alerts

Subscription per desalination plant or fish farm

Agency service

Monitoring contract with environment agencies

Open regional map

Free, builds trust and sampling partners

07 · Datasets

Everything we used is open or published

Source	Used for	Access and licence
NASA PACE OCI Level-2: AOP, BGC, IOP	172-band reflectance, chlorophyll, fluorescence, backscatter	NASA open data, free Earthdata login
GEBCO global bathymetry grid	Shallow-water mask	Free to use with attribution
Kim, Yoo and Son 2016, Optics Express	C. polykrikoides reference, digitised from the paper	Open access, cited
Dwivedi et al. 2015, Environ. Monit. Assess.	Green Noctiluca and diatom references, digitised	Published article, cited
NASA Earth Observatory, Copernicus Image of the Day	Species documentation for the test date	Free to use with credit

Note: Clear water, turbid water and shallow seabed references come from our own PACE scenes.

08 · Methodology

The pipeline runs in six phases

01

Acquire and screen

12 PACE passes. 14 quality flags removed, such as cloud, glint, stray light and steep viewing.

02

Grid and mask

Regridded to 0.01°. Water under 5m depth removed, 5 to 15 m flagged as shallow.

03

Density gate

Dense bloom: chlorophyll of 10 mg/m³ or more AND fluorescence above calm water.

04

Spectral patterns

124 normalised bands, PCA to 5 components (99.2%), then 30 clusters.

05

Name the clusters

Spectral angle to six references over 500 to 700 nm, checked on the derivative.

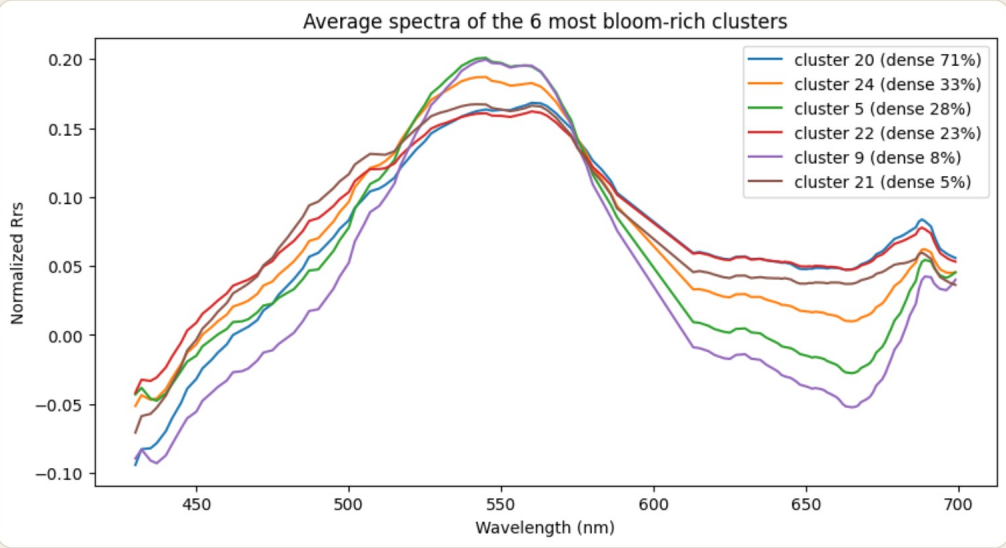
06

Map and export

Label risk tiers and confidence per pixel. Models frozen, then run on the test date.

09 · Naming a Bloom

Each cluster is matched against six reference spectra



The six most bloom-rich clusters. Values below 490 nm turn negative in dense blooms, an atmospheric correction artefact, so naming uses 500 to 700 nm.

Six references

C. polykrikoides (Kim et al. 2016), green *Noctiluca* and diatom (Dwivedi et al. 2015), plus clear, turbid and shallow-seabed water from our scenes.

Two comparisons test

Spectral angle on the normalised spectrum, and again on its smoothed second derivative.

Confidence

(theta_2 minus theta_1) divided by the angle between the two closest references.
A name needs 0.25 or more and both tests agreeing.

10 · Assumptions

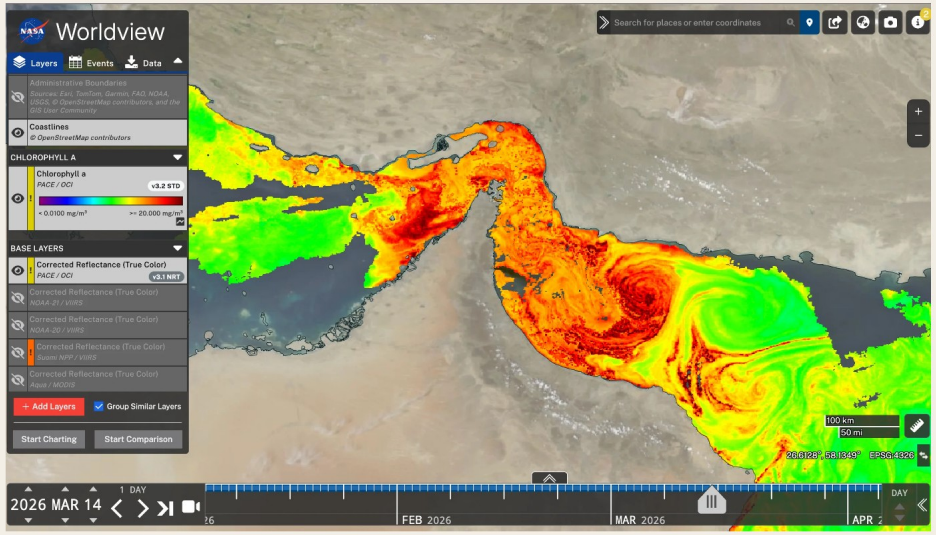
The results rest on six assumptions

1. A cluster's average spectrum stands for its pixels, so each cluster gets one name.
2. Field spectra from Korea and the northern Arabian Sea keep their shape when compared with PACE.
3. The fluorescence threshold set on two calm dates holds in every season.
4. The species signal sits between 500 and 700 nm, where the data stays reliable in dense blooms.
5. NASA and Copernicus are right that the 24 March 2025 bloom was green *Noctiluca*.
6. Water deeper than 5 m is optically deep at 1 km pixels. 5 to 15 m is kept but flagged.

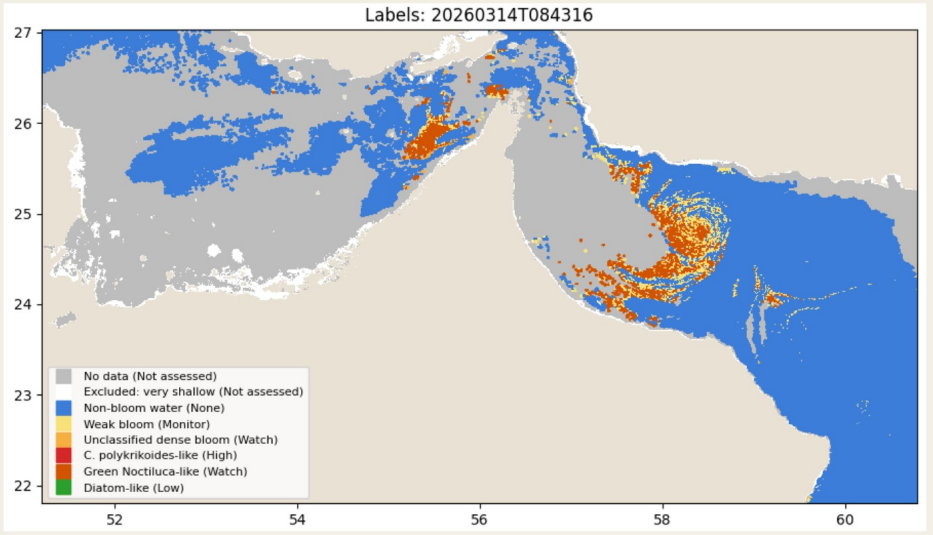
11 · Results: A Large Green Bloom

14 March 2026: the bloom reads as green Noctiluca

NASA Worldview (Chlorophyll layer)



Our Labels

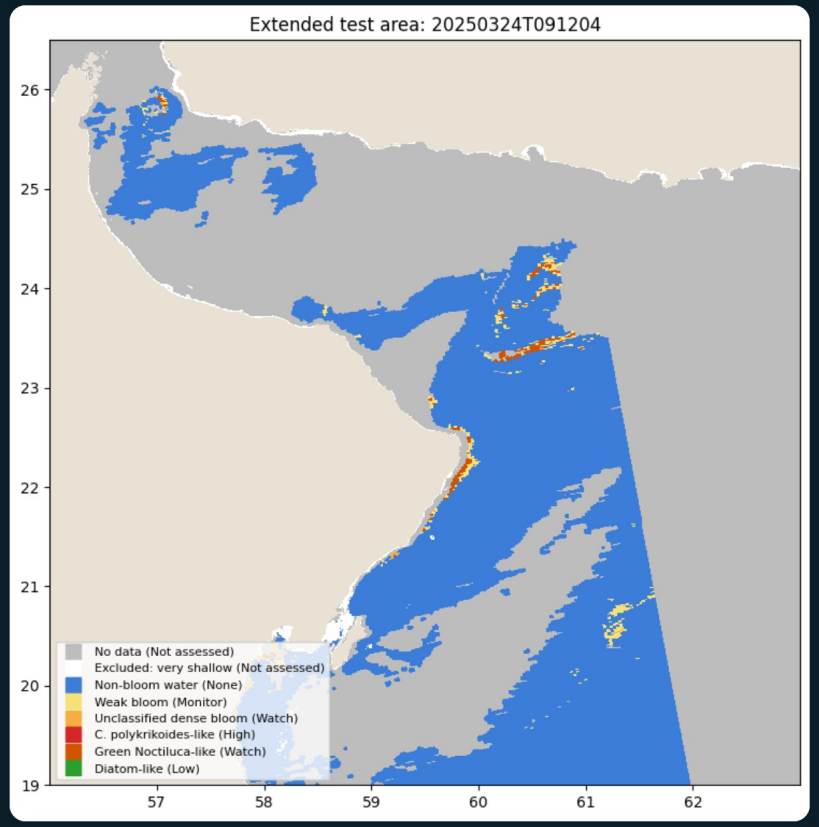


6,718 green *Noctiluca*-like **330** unclassified **0** *C. polykrikoides*-like

Training date: the clusters were fitted on this scene, but species names come only from published reference spectra.

12 · Results: Held-Out Test Date

On 24 March 2025, no pixel was called a red tide



0

pixels labelled *C. polykrikoides*-like
out of 1,220 dense bloom pixels

72.1%

green *Noctiluca*-like (880 px)

27.9%

unclassified (340 px)

NASA and Copernicus both documented the bloom as likely Green Noctiluca for this date but it was not water sampled.

13 · Validation Results: Further Checks

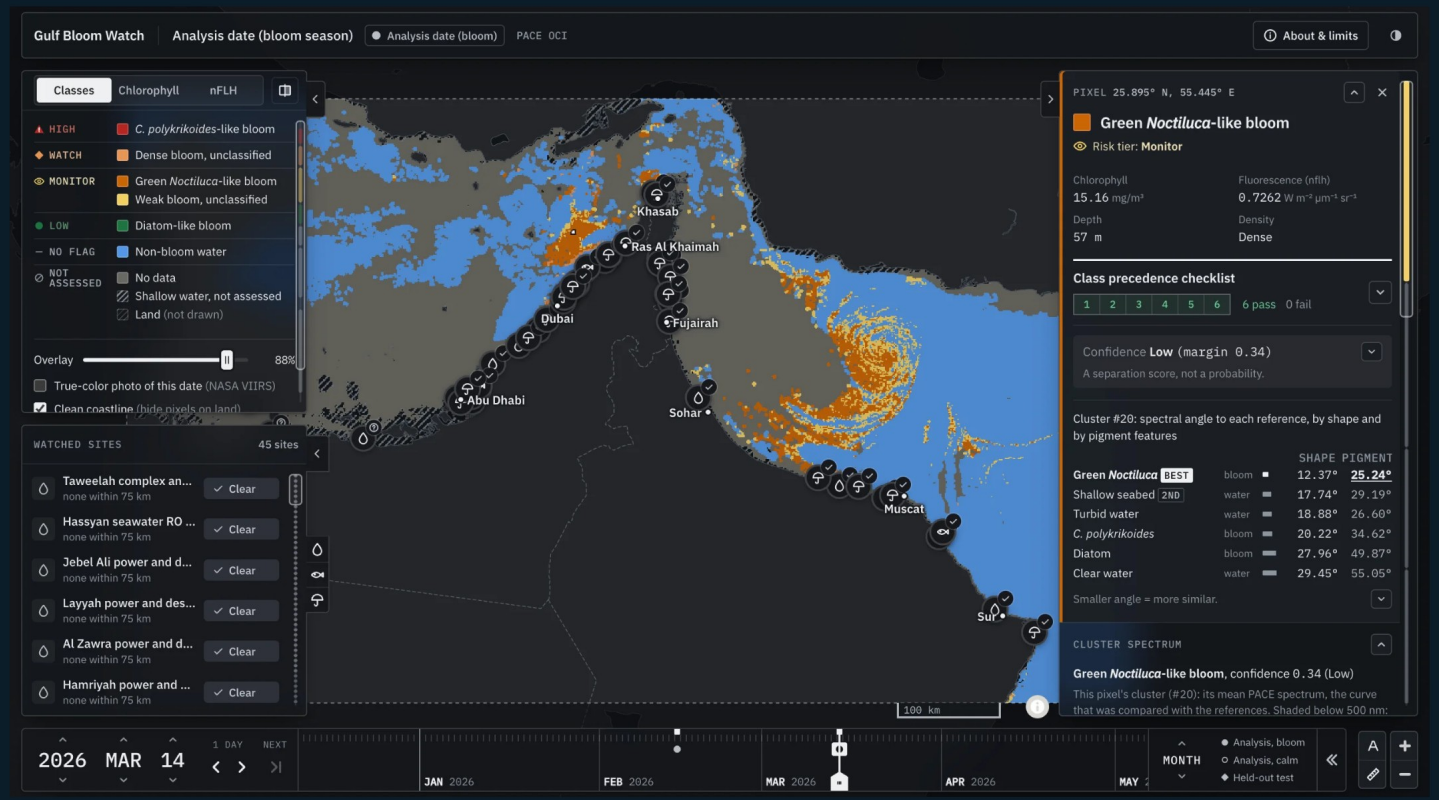
Four more checks point the same way

Check	What we did	Result
Calm dates	Ran on 11 Oct 2024 and 25 Oct 2025	26 dense pixels in total, none called C. polykrikoides
Season and place	Where Noctiluca-like pixels fall, all dates	99% from November to March; 78% on the Sea of Oman side
Synthetic mix	Diluted the C. polykrikoides reference with clear water	Recognised up to about 40% dilution, then unclassified
Band range	Named over 430 to 700 nm instead of 500 to 700	8,803 dense pixels would have been false red tide calls; but with 500 to 700 it gave none

No pixel on any of the 12 dates was called *C. polykrikoides*. This shows the method avoids false alarms. It cannot yet show detection, because PACE has not seen a confirmed red tide here.

14 · Interactive Map

The map is where an operator sees it



Interactive map live at:
https://fireflies-challenge-813.github.io/HAB

15 · Limitations

What this proof of concept cannot show yet

No confirmed red tide (C. polykrikoides) in PACE's record yet, so true detection is untested.

The test bloom's species is documented from imagery, not water sampled.

Species references are digitised from figures, measured in Korea and the Arabian Sea.

Field and PACE spectra of clear water differ by about 13°.

The diatom reference is 4.1° from clear water, so diatom calls are weak.

Bands under 500 nm are dropped because they are over-corrected in dense blooms.

Under 5 m is not assessed and 1 km pixels cannot resolve intake pipes.

Training blooms come mostly from two late-winter 2026 events. Cloud and steep viewing cost some passes.

About 1.5% of offshore moderate-chlorophyll pixels are labelled non-bloom.

16 · Next Steps for Incubation

Incubation would turn our PoC into a service

- Supervised model method once ground truth sampled labels exist
- 30 m hyperspectral from Satellite 813, EnMAP or PRISMA at intakes
- Seabed correction methods for 5 to 15m water
- Gulf-tuned chlorophyll and better correction in dense blooms

- Water sampling with EAD and regional institutes, timed to PACE passes
- PACE-native references from sampled blooms
- More species, such as Trichodesmium and lab diatom spectra
- Automatic run on every new PACE pass in our map
- Alerts to desalination plants and agencies going live

Thank You!

Team FireFlies
Tibyan Khaled Ahmed Bilal
MHD Manar Mallouhi